HAMZA AZIZ

AI Product Engineer | Generative AI & Computer Vision

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Nationality: Indian | UAE employment visa sponsorship required | Notice period: 30 days
Open to relocation: Abu Dhabi (preferred) and Dubai
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PROFESSIONAL SUMMARY

AI Product Engineer with 3+ years of full-time experience developing and optimising production voice AI, LLM applications and computer-vision systems. Maintain and optimise a real-time, multi-tenant DriveThru voice-ordering platform on Google Gemini, with a September baseline reply median of 2.63 s over 17,884 turns and integrations across 4 POS systems; previously delivered facial-recognition systems covering 20+ live CCTV streams with Kafka event streaming. Seeking applied AI engineering roles in Abu Dhabi or Dubai.

WORK EXPERIENCE

AI Product Engineer | *VoicePlug Inc.* | September 2025 – Present

- Maintain and refine a production FastAPI and Socket.IO voice-ordering platform with tenant-scoped workflows, Redis session state and a MongoDB order book, deployed at 2 live restaurants serving 4+ QSR brands; implemented Spanish-language ordering mode
- Optimise the LLM ordering path with conditional Flash-Lite entity extraction before one cached Gemini 2.5 Flash order-agent call, surrounded by deterministic guards, normalisation, rephrase, repeat checks and single-question enforcement; model tool calls are prohibited, with pricing and finalisation handled by application code
- Implemented per-store Vertex AI context caching of the prompt and menu catalogue, Flash/Flash-Lite tiering across five model roles and a 1,024-token reasoning cap to bound long-tail latency; added per-call token-cost accounting
- Helped raise live client success metrics from ~25–30% to 60%+ through prompt optimisation and conversation-flow fixes addressing intent, entity, menu-understanding and fallback failures; converted insights from 500+ real customer interactions into improvements with product, support and QA teams
- Maintain 4 POS integrations (QUBeyond, Clover, Speedline, Deliverect) through a shared adapter with provider-specific pricing, creation, suspension and submission flows; a background per-session cart worker writes schema-bound updates under a Redis lock, and checkout drains the queue before pricing and finalisation
- Fixed cart queue and locking issues; refined entity parsing, size normalisation and combo/multi-size guards to eliminate cart crashes and reject invalid orders before persistence; added fuzzy matching for ASR disfluencies
- Implemented LLM-generated fillers, delayed fillers for stalled turns, a no-LLM pickup fast path and duplicate-TTS suppression when the dialogue response arrives in time
- Built a scripted replay harness checking spoken responses and cart writes, running each case 3 times for model variability, and an ops review loop from production logs through review sheets and replay to prompt/code fixes; improved exception handling, observability and fallback escalation, reducing recurring production issues by 6%

Assistant Engineer – AI | *Global InfoVentures Pvt. Ltd.* | August 2023 – August 2025

- Designed and built the original CCTV attendance application from scratch as its sole developer, integrating third-party InsightFace-REST (SCRFD, w600k_r50) inference with Milvus and achieving >99.99% accuracy against the attendance register in library-gate UAT; delivered 4 CCTV, PTZ and edge-based variants using TensorRT and DeepStream across DGX A100, RTX 4090 and Jetson Orin Nano
- Implemented producer–consumer processing across 20+ live CCTV streams with ROI masking, low-light gating, reconnection and an in-memory frame queue; batched 32 frames across 4 recognition threads using nginx-balanced TensorRT containers on A100 MIG slices, with per-camera Kafka partitions carrying recognition feedback and camera status for Raspberry Pi ID-and-face displays

- Automated face-crop collection and nightly staging-gallery updates from elevated-angle footage, improving real-world recognition accuracy from 85% to 99%; optimised and ported models using ONNX and combined mobile and CCTV datasets, improving cross-domain recognition by 14%
- Built FastAPI ingestion and reporting services with range/list-partitioned PostgreSQL records by date and location; implemented ERP attendance synchronisation scheduled at 1-minute intervals with acknowledgement tracking and manual recovery for a 20-camera main-gate deployment with average daily footfall of ~2,000 people
- Built and deployed a PTZ classroom-attendance system as sole developer, using ONVIF preset sweeps and RTSP capture with faculty captures at lecture start and end, third-party InsightFace-REST inference and Milvus matching, and YOLOv9 headcounts on a GPU MIG slice; stored lecture-wise attendance in year-partitioned PostgreSQL tables, with ERP exports scheduled at 5-minute intervals, success flags and manual resend
- Led a team developing G6, a Dockerised FastAPI voice ERP assistant configured for 4 institutes, with Google speech recognition and gTTS replies; two-stage Rasa routing handles student, employee and fee queries and disambiguates student/employee lookups, achieving near-100% intent classification accuracy using 50+ voice query variations per module
- Led implementation of schema-grounded query generation using 100+ field-level schema mappings per module and per-institute Milvus retrieval of aliases, values and few-shot examples, reducing LLM hallucinations and improving SQL query precision by over 30%; Gemini emits fields and conditions for application-built ERP payloads rather than full SQL; vector-based field/value disambiguation improved SQL generation precision by over 40%

AI Intern | *Global InfoVentures Pvt. Ltd.* | February 2023 – July 2023

- Delivered 10+ AI/ML training sessions, mentored 50+ students across CSE, AIML and Data Science, and guided 10+ YOLOv8/CNN object-detection projects through deployment; built a browser-based PPE violation monitor using a custom YOLOv7 model, Flask, OpenCV and PyTorch, with a 0.75 confidence gate and Gmail API email alerts with cooldown control

SKILLS

- **Generative & Conversational AI:** Google Gemini, Vertex AI, Context Caching, LangGraph, LangChain, Agentic AI, RAG, LLMs, vLLM, Prompt Engineering, Rasa, NLP
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, InsightFace, NumPy, Pandas
- **Inference & Edge AI:** TensorRT, ONNX, DeepStream, NVIDIA Jetson, DGX A100
- **Programming & Backend:** Python, SQL, Bash, TypeScript, FastAPI, Socket.IO
- **Data, Messaging & DevOps:** PostgreSQL, MongoDB, Redis, Milvus (Vector DB), Kafka, MinIO, Docker, CI/CD, Git, Linux

LANGUAGES

- English (Professional Working Proficiency)
- Hindi (Native)

EDUCATION

Master of Technology (Executive), Artificial Intelligence and Machine Learning
BITS Pilani — Work Integrated Learning Programmes (WILP) | 2025 – 2027 (in progress)

Bachelor of Technology, Computer Science and Engineering
ABES Institute of Technology, Dr. A. P. J. Abdul Kalam Technical University (AKTU) | 2019 – 2023

CERTIFICATES

- Rasa Developer Certification — 2025
- Introduction to Data Science, IBM — 2025
- LangChain — Develop AI Agents with LangChain & LangGraph, Udemy — 2025
- Generative AI with Large Language Models, Coursera — 2024

- Building Video AI Applications at the Edge on Jetson Nano, NVIDIA Deep Learning Institute — 2024
- Getting Started with AI on Jetson Nano, NVIDIA Deep Learning Institute — 2024
- Artificial Intelligence and Machine Learning, Global InfoVentures Pvt. Ltd. — 2023
- Getting Started with Deep Learning, NVIDIA Deep Learning Institute — 2022